

Separation of Variables for the Wave Equation on a Circular Drumhead

Physical Problem

We model the vertical displacement $u(r, \theta, t)$ of a circular drumhead of radius 1.

Wave equation:

$$u_{tt} = c^2 \Delta u, \quad 0 \leq r < 1, \quad 0 \leq \theta < 2\pi$$

Boundary condition (fixed edge):

$$u(1, \theta, t) = 0$$

Initial conditions:

$$u(r, \theta, 0) = u_0(r, \theta), \quad u_t(r, \theta, 0) = v_0(r, \theta)$$

The Laplacian in Polar Coordinates

In polar coordinates,

$$\Delta u = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}$$

Thus the wave equation becomes

$$u_{tt} = c^2 \left(u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} \right)$$

The circular geometry suggests separation of variables in r , θ , and t .

Separation of Variables Ansatz

Assume a product solution

$$u(r, \theta, t) = R(r) \Theta(\theta) T(t)$$

Substitute into the PDE and divide by $R\Theta T$:

$$\frac{T''}{c^2 T} = \frac{R''}{R} + \frac{1}{r} \frac{R'}{R} + \frac{1}{r^2} \frac{\Theta''}{\Theta}$$

$$\frac{T''}{c^2 T} = \lambda \quad \frac{\Theta''}{\Theta} = \mu \quad \frac{R''}{R} + \frac{1}{r} \frac{R'}{R} + \frac{\mu}{r^2} = \lambda$$

Angular Eigenvalue Problem

The angular equation is

$$\Theta'' + m^2\Theta = 0$$

Periodicity condition:

$$\Theta(\theta + 2\pi) = \Theta(\theta)$$

Eigenvalues and eigenfunctions:

$$m = 0, 1, 2, \dots$$

$$\Theta_m(\theta) = \begin{cases} 1, & m = 0, \\ \cos(m\theta), \sin(m\theta), & m \geq 1 \end{cases}$$

Radial Eigenvalue Problem

The radial equation becomes

$$r^2 R'' + rR' + (\lambda^2 r^2 - m^2)R = 0$$

This is **Bessel's equation** of order m .

Regularity at $r = 0$ excludes singular solutions, so

$$R(r) = J_m(\lambda r)$$

The boundary condition $R(1) = 0$ gives

$$J_m(\lambda) = 0$$

Bessel Zeros and Eigenvalues

Let $\alpha_{m,k}$ denote the k -th positive zero of J_m .

Radial eigenvalues:

$$\lambda_{m,k} = \alpha_{m,k}^2$$

Radial eigenfunctions:

$$R_{m,k}(r) = J_m(\alpha_{m,k}r)$$

Each pair (m, k) gives a vibrational mode of the drumhead.

Time Equation

The time ODE is

$$T'' + c^2 \lambda_{m,k}^2 T = 0$$

Solutions:

$$T_{m,k}(t) = A_{m,k} \cos(c\alpha_{m,k}t) + B_{m,k} \sin(c\alpha_{m,k}t)$$

Normal Modes

Each normal mode has the form

$$u_{m,k}(r, \theta, t) = J_m(\alpha_{m,k} r) [C \cos(m\theta) + D \sin(m\theta)] [A \cos(c\alpha_{m,k} t) + B \sin(c\alpha_{m,k} t)]$$

The general solution is a superposition of all such modes.

General Solution

$$u(r, \theta, t) = \sum_{m=0}^{\infty} \sum_{k=1}^{\infty} \left[a_{m,k} \cos(c\alpha_{m,k}t) + b_{m,k} \sin(c\alpha_{m,k}t) \right] v_{m,k}(r, \theta)$$

where

$$v_{m,k}(r, \theta) = J_m(\alpha_{m,k}r) \Phi_m(\theta)$$

$$\Phi_m(\theta) = \begin{cases} 1, & m = 0, \\ \cos(m\theta), \sin(m\theta), & m \geq 1 \end{cases}$$

Determining the Coefficients

Coefficients are determined from the initial data:

$$u(r, \theta, 0) = u_0(r, \theta), \quad u_t(r, \theta, 0) = v_0(r, \theta)$$

Expand u_0 and v_0 in the orthogonal basis

$$J_m(\alpha_{m,k}r) \cos(m\theta), \quad J_m(\alpha_{m,k}r) \sin(m\theta)$$

using radial and trig orthogonality

$$\int_0^1 r J_m(\alpha_{m,k}r) J_m(\alpha_{m,\ell}r) dr = 0 \quad (k \neq \ell)$$

Understanding Nodal Patterns

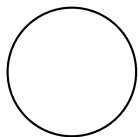
For each mode (m, k) :

- ▶ m : number of **nodal diameters** (angular nodal lines)
- ▶ $k - 1$: number of **nodal circles** (radial nodal lines)

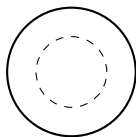
Key ideas:

- ▶ Nodal lines remain stationary
- ▶ Regions between nodal lines oscillate with alternating sign
- ▶ Higher modes correspond to higher frequencies

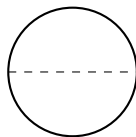
Low-Order Nodal Patterns (Schematics)



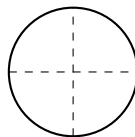
$m, k = 0, 1$



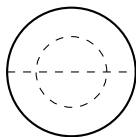
$m, k = 0, 2$



$m, k = 1, 1$



$m, k = 2, 1$



$m, k = 1, 2$

Dashed lines indicate nodal diameters and nodal circles.

Physical Interpretation

- ▶ Each (m, k) is a vibrational mode of the membrane
- ▶ m : number of angular nodal lines
- ▶ k : number of radial nodes (including the boundary)
- ▶ Frequencies:

$$\omega_{m,k} = c \alpha_{m,k}$$

- ▶ Different modes oscillate independently

Summary

- ▶ The wave equation on a disk separates naturally in polar coordinates
- ▶ Angular dependence gives Fourier modes
- ▶ Radial dependence leads to Bessel functions
- ▶ Boundary conditions select Bessel zeros
- ▶ Initial displacement and velocity determine modal amplitudes